

COMP7940 Cloud Computing

Lab 3: Consuming External Service – ChatGPT

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1. Architecture of the Chatbot System

The chatbot system consists of the following components:

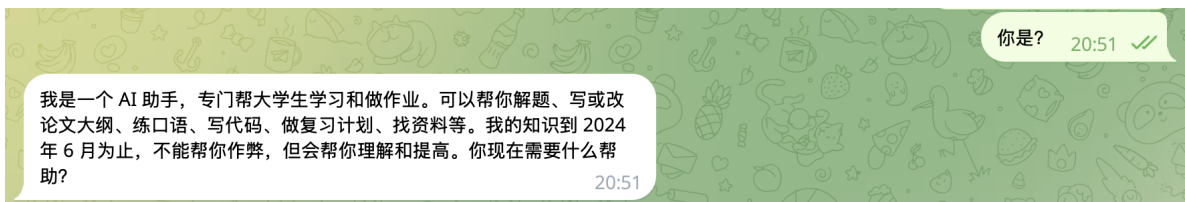
- **Telegram Bot Client:** Runs on the user's local machine (or server). It uses `python-telegram-bot` [3, 4] to connect to Telegram's API and receive/send messages via long polling.
- **chatbot.py:** The main entry point running locally. It loads `config.ini`, creates a ChatGPT client, registers message handlers, and starts polling.
- **ChatGPT_HKBU.py:** A local Python module that wraps the HKBU ChatGPT REST API. It reads API configuration from `config.ini` and sends HTTP POST requests to the HKBU GenAI endpoint.
- **HKBU GenAI API** (<https://genai.hkbu.edu.hk/api/v0/rest>): A remote cloud service hosted by HKBU [1, 2]. It receives chat completion requests and returns LLM-generated responses.

Where components run:

- Local: `chatbot.py`, `ChatGPT_HKBU.py`, `config.ini`
- Remote: Telegram servers (message delivery), HKBU GenAI API (LLM inference)

2. Screenshots of Telegram Bot with ChatGPT

Here are some screenshots of the Telegram bot with ChatGPT.



What is SNP? 20:51 ✓

Do you mean SNP in genetics or the Scottish National Party? I'll explain the genetics meaning first, since that's the most common usage in university science classes — tell me if you meant the party instead.

SNP (single nucleotide polymorphism) — simple explanation:

- It's a change of a single DNA base (A, T, C, or G) at a specific position in the genome.
- Example: most people might have an A at position X, while some have a G there — that single-base difference is an SNP.
- By convention, something is called an SNP if the less common version (the minor allele) appears in at least about 1% of the population.

Key points:

- SNPs are the most common form of genetic variation in humans — roughly one SNP every 1,000 bases.
- They can be in coding regions (genes), noncoding regions (between genes), or regulatory regions (promoters/enhancers).
- Effects:
 - No effect (synonymous change or in nonfunctional DNA).
 - Change the protein sequence (nonsynonymous) and possibly change function.
 - Affect gene regulation (how much or when a gene is expressed).
- Genotypes: people can be homozygous (same allele on both chromosomes) or heterozygous (two different alleles).
- Uses in research and medicine:
 - Genome-wide association studies (GWAS) to find SNPs linked to diseases or traits.
 - Population genetics and ancestry analysis.
 - Pharmacogenomics (how people respond to drugs).
 - Forensics and genetic fingerprinting.

How they're detected:

- Genotyping arrays (chips) that test many known SNPs at once.
- DNA sequencing (whole-genome or targeted) which can discover new SNPs.

Databases and naming:

- Public databases like dbSNP catalog known SNPs; common IDs look like "rs123456".

If you meant the political SNP (Scottish National Party) I can give a short summary of that instead. Which one did you want?

20:51 ❤️

3. Terminal Logs

Here are the logs from running `python chatbot.py`:

```
(comp7940-lab) zhaozhan@zhaozhan-macbookair lab3 % export HTTPS_PROXY=http://127.0.0.1:7897
python chatbot.py
2026-02-23 20:51:10,153 - root - INFO - INIT: Loading configuration...
2026-02-23 20:51:10,153 - root - INFO - INIT: Connecting the Telegram bot...
2026-02-23 20:51:10,153 - root - INFO - INIT: Using proxy: http://127.0.0.1:7897
2026-02-23 20:51:10,211 - root - INFO - INIT: Registering the message handler...
2026-02-23 20:51:10,211 - root - INFO - INIT: Initialization done!
2026-02-23 20:51:12,939 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/getMe "HTTP/1.1 200 OK"
2026-02-23 20:51:13,206 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/deleteWebhook "HTTP/1.1 200 OK"
2026-02-23 20:51:13,208 - telegram.ext.Application - INFO - Application started
2026-02-23 20:51:18,213 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/getUpdates "HTTP/1.1 200 OK"
2026-02-23 20:51:18,217 - root - INFO - UPDATE: Update(message=Message(channel_chat_created=False, chat=Chat(first_name='Zhao', id=
8344683154, last_name='Zhan', type=<ChatType.PRIVATE>, username='Extious'), date=datetime.datetime(2026, 2, 23, 12, 51, 17, tzinfo=
datetime.timezone.utc), delete_chat_photo=False, from_user=User(first_name='Zhao', id=8344683154, is_bot=False, language_code='zh-h
ans', last_name='Zhan', username='Extious'), group_chat_created=False, message_id=200, supergroup_chat_created=False, text='你是? '
), update_id=274769347)
2026-02-23 20:51:19,687 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/sendMessage "HTTP/1.1 200 OK"
2026-02-23 20:51:28,480 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/getUpdates "HTTP/1.1 200 OK"
2026-02-23 20:51:28,619 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/editMessageText "HTTP/1.1 200 OK"
2026-02-23 20:51:38,741 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/getUpdates "HTTP/1.1 200 OK"
2026-02-23 20:51:44,421 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/getUpdates "HTTP/1.1 200 OK"
2026-02-23 20:51:44,424 - root - INFO - UPDATE: Update(message=Message(channel_chat_created=False, chat=Chat(first_name='Zhao', id=
8344683154, last_name='Zhan', type=<ChatType.PRIVATE>, username='Extious'), date=datetime.datetime(2026, 2, 23, 12, 51, 43, tzinfo=
datetime.timezone.utc), delete_chat_photo=False, from_user=User(first_name='Zhao', id=8344683154, is_bot=False, language_code='zh-h
ans', last_name='Zhan', username='Extious'), group_chat_created=False, message_id=202, supergroup_chat_created=False, text='What is
SNP?'), update_id=274769348)
2026-02-23 20:51:45,908 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/sendMessage "HTTP/1.1 200 OK"
2026-02-23 20:51:54,688 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/getUpdates "HTTP/1.1 200 OK"
2026-02-23 20:51:55,031 - root - WARNING - Telegram network error, retry in 1.5s: httpx.ConnectError:
2026-02-23 20:51:58,154 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/editMessageText "HTTP/1.1 200 OK"
2026-02-23 20:52:04,952 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/getUpdates "HTTP/1.1 200 OK"
2026-02-23 20:52:15,216 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKisR
cIA/getUpdates "HTTP/1.1 200 OK"
^C2026-02-23 20:52:20,632 - httpx - INFO - HTTP Request: POST https://api.telegram.org/bot8392033952:AAEI_smXr20T6GrJR-r8yeYWw_bKKi
sRcIA/getUpdates "HTTP/1.1 200 OK"
```

References

- [1] Hong Kong Baptist University. Hkbu genai platform. Available at <https://genai.hkbu.edu.hk/>, 2026. Accessed: 2026-02-23.
- [2] OpenAI. Openai api documentation. Available at <https://platform.openai.com/docs/>, 2026. Accessed: 2026-02-23.
- [3] python-telegram-bot Community. python-telegram-bot documentation. Available at <https://docs.python-telegram-bot.org/>, 2026. Accessed: 2026-02-23.
- [4] Telegram. Telegram bot api. Available at <https://core.telegram.org/bots/api>, 2026. Accessed: 2026-02-23.